UU-T-118B
May 18, 1976
SUPERSEDING
Int. Fed. Spec. UU-T-00118A(DSA-DM)
May 4, 1966 and
Fed. Spec. UU-T-118
February 12, 1959

FEDERAL SPECIFICATION

TAPE, SEALING, STERILIZATION INDICATOR

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers one type of creped paper-backed tape, suitable for use in sealing or holding surgical pack wrappers, or for application to stainless steel, enamel, porcelain, or glass, to indicate that these items have been subjected to prolonged exposure to high temperature encountered in steam sterilization in hospital autoclaves.

1.2 Classification. Tape covered by this specification shall be of the following sizes, as specified (see 6.2).

Size 1 - 1/2 inch wide by 60 yards long
Size 2 - 1 inch wide by 60 yards long

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

Federal Specification:

PPP-B-636 - Boxes, Shipping, Fiberboard

Federal Standards:

Fed. Std. No. 66 - Steel: Chemical Composition and Hardenability
Fed. Test Method Std. No. 101 - Preservation, Packaging, and Packing Materials, Test Procedures
Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

FSC 6530

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(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.

Military Standards:

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The tape shall consist of creped paper, coated on one side with a pressure-sensitive, water-insoluble adhesive requiring no moisture, heat, or other special preparation prior to application.

3.2 Tape. The tape shall be crepe embossed, uniform in texture, and shall have sufficient internal strength to allow the tape to be unwound from the roll, to hold the surgical pack closed during autoclaving, and to be removable from the pack after autoclaving, without breakage or delamination. It shall be possible to make pencil and permanent ink markings on the paper backing of the tape, and such markings shall be legible after autoclaving as specified in 4.4.8. The tape backing shall be so treated in the manufacturing process that after standard steam autoclaving (250°F., 20 minutes), clearly visible identifying markings shall appear on the backing so as to indicate that the tape has been subjected to autoclaving conditions. Marking treatment used for this purpose shall be activated only by the conditions of the standard autoclaving process. Markings, which appear after autoclaving, shall be in the form

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of well-defined lines darker than the tape backing. When tested as specified in 4.4.6.2, there shall be no visible change in the color of the autoclave indicator. When tested as specified in 4.4.7, suspension of the test panel for ten seconds or longer shall constitute failure and shall be cause for rejection of the tape. When tested as specified in 4.4.8.1, there shall be a visible color change in the autoclave indicator ink. When tested as specified in 4.4.8.2, there shall be no visible color change in the autoclave indicator. When tested as specified in 4.4.8.3, there shall be no definite ink stains on the cloth conforming to the indicator ink pattern on the test tape.

3.3 Adhesive. The adhesive shall be such that the tape shall adhere immediately and firmly to clean, dry surfaces to which the tape is applied. During autoclaving operations, there shall be no wrinkling, curling, breaking, or lifting of the tape, such as to make the tape unfit for use. A slight wrinkling, curling, breaking, or lifting of the tape edge shall be acceptable. After autoclaving, the tape shall be capable of being removed, without breakage and without transfer or splitting of the adhesive. There shall be no separation of the adhesive from the backing upon unwinding the rolls. There shall be no liner over the adhesive.

3.4 Rolls. The tape shall be uniformly and smoothly wound in rolls, with the adhesive side in, on suitable cores having an inside diameter of 3 ± 1/32 inches. The cores shall have sufficient rigidity to prevent distortion of the roll under normal conditions of transportation, storage, and use. The tape shall be one continuous strip, except that any single roll may contain two splices. Any splices shall be such that they will not separate when the roll is unwound in test or in application. The standard minimum length of tape in the roll shall be 60 yards. The width of the tape shall be 1/2 inch for size 1 and 1 inch for size 2. A tolerance of ± 1/8 inch shall be allowed for size 1 and size 2 tape widths.

3.5 Physical properties. When tested as specified in section 4, the averages of the test values for the tapes shall conform to the requirements specified in table I.

TABLE I. Physical properties

Property	Requirement	Test paragraph
Tensile breaking strength (pounds per inch width), minimum:		
Wet	4.5	4.4.3
Dry	16	4.4.4
Ultimate elongation (percent), minimum	5	4.4.4.1
Thickness (inch), maximum	0.008	4.4.5

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TABLE I. Physical properties (continued)

Property	Requirement	Test paragraph
Adhesion to steel (ounces per inch width), minimum:		
Initial	30	4.4.6.1
After humidity test	30	4.4.6.2
After accelerated aging test	22	4.4.6.3
Autoclave adhesive stability test, (seconds), maximum	10	4.4.7
Autoclave indicator ink tests		
Steam autoclave test	Clearly visible color change	4.4.8.1
Dry humidity test	No color change	4.4.8.2
High humidity test	No color change	4.4.6.2
Cloth staining test	No cloth stain	4.4.8.3

3.6 Workmanship. Workmanship shall be first class throughout. Tapes shall be free from defects which detract from their appearance or may impair their serviceability. The adhesive shall be a smooth, uniform coating, covering the entire area of one side of the tape. Edges of the tape shall be straight, true, and unbroken.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

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4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the tape may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table II.

TABLE II. Sampling for examination

	Inspection level	AQL (defects per hundred units)
For visual examination		
Major defects	II	1.0
Total (major and minor defects combined)	II	4.0
Dimensional examination	S-3	2.5

4.2.2 For tests. For lots of 8000 rolls, or less, 10 rolls shall be selected at random for tests. For lots larger than 8000 rolls, 15 rolls shall be selected for tests. Acceptance number shall be zero in either case. Where possible, the same sample shall be used for the determination of more than one test.

4.3 Examination. The tape shall be examined to determine compliance with requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the classification of defects, shown in table III. Examination shall not be restricted to the classified possible defects listed below.

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TABLE III Classification of defects

Categories	Defects
Major	
101	Tape rolls not smoothly or uniformly wound.
102	Sides of tape not free of cuts and deformed areas.
103	Roll coned; roll telescopes when subject to flexure caused by own weight.
104	Adhesives separate from backing upon unwinding roll.
105	Tape not crepe embossed.
106	Intermittent breaks in adhesive coating.
107	Tape not free of holes, tears, breaks, cuts and splits.
108	Adhesive coating on roll not uniform and free of breaks.
109	Number of splices in roll not within the maximum allowed.
Minor	
201	Tape roll crushed to the extent of presenting an oval rather than round profile.
202	Inside diameter of roll cores not within tolerance specified.
203	Rolls and tape not free of adhering or imbedded foreign material, dirt, fragments of paper and excess adhesives.

4.3.2 Dimensional examination. The tape shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Specimens. Three specimens for each test shall be taken from each sample roll. The first three laps of tape shall be removed and specimens for testing shall then be taken from the beginning and two other areas near the end and near the midpoint of the roll. The average of the three values obtained for each test on each roll shall be used to determine compliance with requirements.

4.4.2 Conditioning. The tape shall be conditioned for not less than 24 hours at 73°F ± 3.5° F. and 50 ± 2 percent relative humidity, and shall be tested under these conditions.

4.4.3 Tensile breaking strength, wet. Samples of tape, 11 inches in length, shall be immersed in boiling water for 5 minutes and then immediately transferred to a container of water at 73°F ± 3.5°F., where they shall remain until tested. Testing shall be completed within 10 minutes. The sample to be tested shall be removed from the water and placed, adhesive side up, on paper toweling for not more than 5 seconds, to absorb excess water.

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4.4.4 Tensile breaking strength, dry. The procedure for testing the tensile breaking strength of dry tape shall be as described in Federal Test Method Standard 101, Method No. 2062.

4.4.4.1 Ultimate elongation, dry. The procedure for testing the ultimate elongation of dry tape shall be as described in Federal Test Method Standard 101, Method No. 2062.

4.4.5 Thickness. The thickness of the tape shall be determined in accordance with Federal Test Method Standard 101, Method No. 1004.

4.4.6 Adhesion test.

4.4.6.1 Adhesion to steel (initial). The tape shall be tested to determine compliance with requirements specified in table I by testing as per Federal Test Method Standard 101, Method No. 2050.

4.4.6.2 Humidity test. Rolls with 10 ± 1 yards of tape shall be suspended in a vertical position over water in a sealed container for 7 days at 73°F ± 3.5° F. At the end of this period, the tape shall be removed and immediately tested for adhesion to steel as specified in 4.4.6.1 When the tape is unwound at 6 to 8 inches per second, there shall be no adhesive transfer or splitting and the backing shall not break. Examine the outer laps of tape from each roll for color change. The tape shall comply with the requirements specified in table I.

4.4.6.3 Accelerated aging test. Accelerated aging of tapes is performed as specified in Federal Test Method Standard 101, Method No. 4047, Procedure A, except that the tape samples are exposed to heat and humidity for 12 days before testing. After removal from accelerated aging apparatus, test rolls are allowed to cool for 2 hours, and are then tested for adhesion as specified in 4.4.6.1. When tapes are unwound at 6 to 8 inches per second, there shall be no adhesive transfer or splitting, and the backing shall not break. The tape shall comply with requirements specified in table I.

4.4.7 Autoclaving adhesive stability test. The tape shall be tested to determine the stability of the adhesive after autoclaving. Samples of tape, 6-1/2 inches in length, shall be rolled into contact with cleaned test panels covering a surface area of 1 inch by 5 inches on each panel. The extending 1-1/2 inch of the tape shall be folded adhesive to adhesive to form a tab. The test assemblies shall be placed horizontally in a steam autoclave at a temperature of 250°F + 5°F - 2° F. for 30 minutes. After this period the steam shall be shut off, the exhaust vents opened and the samples allowed to cool in the autoclave for 30 minutes. After removal, the assemblies shall be allowed to air-dry for 4 hours and then the tape shall be peeled from the test panels at a 180° angle and at a rate of 10 to 12 inches per second. A clean

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2 inches by 5 inches sheet of 0.003 inch thick dead soft aluminum foil shall be rolled onto the test panel and into intimate contact with any adhesive residue. The roller shall be placed initially at the center of the panel and then moved back and forth at a rate of 5 to 7 inches per second. The foil shall be traversed lengthwise a total of four times. A strip of tape shall be banded with its adhesive surface exposed, around a smoothly finished block of wood. The block of wood shall be 5 inches long, 2 inches wide, and sufficiently thick to provide a total weight of 0.33 pound ± 0.05 pound. The first inch of the tape shall be adhered to the upper surface of the block and the rest shall be folded at a 180° angle, encircling the block in a direction such that the length of the tape is parallel to the 5 inch length of the block. The block shall be centered on the test panel such that the tape contacts the entire surface of the aluminum foil. No downward pressure shall be exerted on the block. After contact for 3 seconds the block shall be slowly lifted and the test panel observed for conformance to requirements specified in 3.2. The tape shall comply with requirements specified in table I.

4.4.8 Autoclave indicator ink test. The three test samples shall be prepared and tested to determine compliance with requirements of table I. A 2 inches by 7-1/2 inches strip of white surgical pack wrapper cloth shall be applied to each of three steel test panels. The cloth shall be secured by taping the ends to the test panels. A 1 inch by 3 inch strip of test tape shall be centered on each cloth covered panel. The test panels shall be used in the tests specified in 4.4.8.1, 4.4.8.2, and 4.4.8.3.

4.4.8.1 Steam autoclave test. One of the panels with the backing exposed shall be autoclaved for 30 minutes at 250° + 5° -2°F. and then allowed to dry for 30 minutes in the autoclave with steam pressure in the jacket and the door open. After removal from the autoclave the samples shall be allowed to cool for 30 minutes and then examined for color change. The tape shall comply with requirements specified in table I.

4.4.8.2 Dry heat test. One panel, with tape backing exposed, shall be placed in a dry heat oven, for 30 minutes at 250° + 5° - 2°F. The panel shall be removed and allowed to cool for 30 minutes and then examined for color change. The tape shall comply with requirements specified in table I.

4.4.8.3 Cloth staining test. The tape shall be tested to determine compliance with requirements of 3.2. A 1-1/2 inch by 4 inch strip of white surgical pack wrapper cloth shall be centered over the tape on one of the panels prepared as specified in 4.4.8. The cloth shall be taped to insure intimate contact between the cloth and the indicator ink on the test tape. The test panel shall be placed, cloth side up, in a steam autoclave for 30 minutes at 250° + 5° -2°F. and then allowed to dry in the autoclave with steam pressure in the jacket and the door open. The test panel shall be removed and allowed to cool for 30 minutes. The upper layer of cloth shall be removed and examined for ink stains. The tape shall comply with requirements specified in table I.

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4.5 Examination for preparation for delivery. An examination shall be made to determine whether the packaging, packing,and marking comply with the requirements of section 5. Defects shall be scored as specified in table IV. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 defects per hundred units.

TABLE IV. Examination for preparation for delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Packaging and Packing Materials	Any component missing or damaged.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper size, location, sequence, or method of application.

5. PACKAGING

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Unit of issue. One roll containing sixty (60) yards of tape of the size specified, constitutes one unit of issue.

5.1.1.1 Level A.

5.1.1.1.1 Unit package. Each roll of tape shall be packaged in a container of suitable size and design which will permit dispensing of the tape without removing roll from the container. Container shall be provided with a metal cutting edge with teeth, located in a position such as to permit cutting the tape while dispensing. Container design shall include a centering device or fiberboard ring to prevent contact between the outside lap of the tape roll and the inner face of the container.

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5.1.1.1.2 Intermediate package. Twelve units of issue shall be packaged in a box of appropriate size constructed in accordance with PPP-B-636, type CF, class domestic. Closure shall be adequate to prevent accidental opening.

5.1.1.2 Level C. Tape shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. One hundred forty-four units of issue of size 1 or seventy-two (72) units of issue of size 2 shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, class weather resistant, grade V3c. Closure, waterproofing and strapping shall be as specified in the appendix of PPP-B-636.

5.2.2 Level B. One hundred forty-four units of issue of size 1 or seventy-two (72) units of issue of size 2 shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, type CF, class domestic. Closure shall be as specified in the appendix of PPP-B-636.

5.2.3 Level C. Tape shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Civil agencies. Marking for shipment shall be in accordance with Fed. Std. No. 123.

5.3.2 Military activities. Each unit, intermediate package and exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. To provide a necessary sealing tape for use in sealing or holding surgical pack wrappers, or for application to clinical utensils and medical glass items, during prolonged exposure to the high temperatures and pressures encountered in steam sterilization in hospital autoclaves.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

(a) Title, number and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(c) Size required (see 1.2).

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6.3 This specification covers the following items appearing in the Federal Supply Catalog:

National Stock Number	Item Identification
6530-00-299-9821	TAPE, SEALING, STERILIZATION INDICATOR, Steam, Pressure Sensitive Adhesive, 1 inch wide by 60 yards long.
6530-00-299-9822	TAPE, SEALING, STERILIZATION INDICATOR, Steam, Pressure Sensitive Adhesive, 1/2 inch wide by 60 yards long.

MILITARY CUSTODIANS

Army - MD
Navy - MS
Air Force - 03

Preparing activity:

DSA-DM

Civil Agency Coordinating Activities:

VA-DMS
HEW-HSM
NASA-JFK
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